Surds

Questions 1–26

Question 1. We would expect completion of these questions without the use of a calculator. Simplify

a) $\sqrt{32}$
b) $2\sqrt{12}$
c) $\sqrt{45}$
d) $\sqrt{200}$
e) $-4\sqrt{50}$

Question 2. Simplify

a) $\sqrt{5} \times \sqrt{2}$
b) $\sqrt{5}\sqrt{15}$
c) $\sqrt{8} \times 3\sqrt{2}$
d) $\sqrt{\frac{1}{9}}$
e) $\frac{\sqrt{28}}{\sqrt{14}}$
f) $\sqrt{25x^2}$

Question 3. Simplify

a) $2\sqrt{5} - \sqrt{5}$
b) $\sqrt{8} + 2\sqrt{2}$
c) $\sqrt{12} - \sqrt{3}$
d) $\sqrt{128} - \sqrt{72}$
e) $3\sqrt{12} + \sqrt{27}$
f) $\sqrt{150} - \sqrt{96}$
g) $\sqrt{3} + \sqrt{27} + \sqrt{12}$
h) $\sqrt{350} + \sqrt{24}$
i) $\sqrt{32a} + 2\sqrt{a} - \sqrt{28a}$

Question 4. Simplify

a) $\sqrt{2} \times \sqrt{7} \times \sqrt{3}$
b) $\sqrt{18} \times \sqrt{2}$
c) $2\sqrt{3} \times 2\sqrt{3}$
d) $5\sqrt{3} \times 2\sqrt{5}$
e) $-3\sqrt{3} \times 2\sqrt{3}$
f) $(2\sqrt{6})^2$
g) $(3\sqrt{8})^2$
h) $(2\sqrt{5})^3$

Question 5. Express each of the following in the form $\sqrt{a}$ where a is a positive integer:

a) $2\sqrt{5}$
b) $4\sqrt{3}$
c) $3\sqrt{2}$
d) $2\sqrt{7}$
e) $2\sqrt{43}$

Question 6. Express the following as a single square root:

a) $a\sqrt{b}$
b) $2\sqrt{a}$
c) $(\sqrt{a})^5$
d) $(a\sqrt{b})^3$

Question 7. Express $\sqrt{525}$ in the form $p\sqrt{q}\sqrt{r}$ where p, q and r are prime numbers.

Question 8. Write this ratio in its simplest form: $\sqrt{300} : \sqrt{48} : \sqrt{27}$.

Question 9. Rationalise the following surds:

a) $\frac{3}{\sqrt{2}}$
b) $\frac{6}{\sqrt{3}}$
c) $\frac{6}{\sqrt{10}}$
d) $\frac{2}{\sqrt{6}}$
e) $\frac{10}{\sqrt{10}}$
f) $\frac{4\sqrt{2}}{\sqrt{8}}$

Question 10. Expand and simplify the following:

a) $\sqrt{5}(1+\sqrt{5})$
b) $\sqrt{11}(2+\sqrt{11})$
c) $4(\sqrt{3}+2\sqrt{5})$
d) $(\sqrt{3}+1)(\sqrt{3}-1)$
e) $(\sqrt{6}+1)(2+\sqrt{6})$
f) $(\sqrt{8}-1)(2+\sqrt{8})$
g) $(\sqrt{24}+1)(\sqrt{2}-3)$

Question 11. Rationalise the following surds:

a) $\frac{1}{\sqrt{2}+1}$

b) $\frac{3}{\sqrt{5}-2}$

c) $\frac{2}{1-\sqrt{6}}$

d) $\frac{1}{a+\sqrt{b}}$

e) $\frac{1}{a-\sqrt{2b}}$

Question 12. Simplify the following, giving your answers in the form $p\sqrt{q}$ where p and q are non-zero rational numbers:

a) $\sqrt{27}+\frac{15}{\sqrt{3}}$

b) $\frac{\sqrt{54}}{2}-\frac{10}{\sqrt{6}}$

c) $\frac{\sqrt{150}}{5}+\frac{30}{\sqrt{24}}$

Question 13. Solve $x(4-\sqrt{6})=10$ giving your answer in the form $a+b\sqrt{6}$, where a and b are non-zero integers to be found.

Question 14. Rationalise the denominator and simplify the following:

a) $\frac{\sqrt{3}}{\sqrt{2}-2}$

b) $\frac{2\sqrt{5}-1}{\sqrt{5}+4}$

c) $\frac{5\sqrt{3}-2}{2-\sqrt{7}}$

d) $\frac{2\sqrt{3}-7}{2\sqrt{3}-3}$

e) $\frac{1-\sqrt{k}}{2+\sqrt{2k}}$

Question 15. Solve $x(1-\sqrt{2})+\sqrt{2}=5$ giving your answer in the form $a+b\sqrt{2}$, where a and b are non-zero integers to be found.

Question 16. Show that $\frac{3\sqrt{10}}{\sqrt{2}+\sqrt{5}}$ may be written in the form $p\sqrt{q}-q\sqrt{p}$ where p and q are non-zero integers to be found.

Question 17. Show that $\frac{\sqrt{3}+\sqrt{7}}{\sqrt{3}-\sqrt{7}}$ may be written in the form $a+b\sqrt{21}$ where a and b are non-zero constants to be found.

Question 18. The expression $\frac{a+\sqrt{6}}{1-\sqrt{6}}$ may be written in the form $p+q\sqrt{6}$. Find the values of p and q in terms of a.

Question 19. Fully simplify the expression $\frac{12}{\frac{1}{\sqrt{3}}+\sqrt{3}}$.

Question 20. A cylinder has a height of $(\sqrt{3}+1)$cm and a radius of $(\frac{1}{\sqrt{3}-2})$cm. Show that the volume of the cylinder is exactly $\pi(19+11\sqrt{3})\text{cm}^3$.

Question 21. The expression $\frac{-k+2\sqrt{3}}{\sqrt{27}+1}$ may be written in the form $a+b\sqrt{3}$. Find the values of a and b in terms of k.

Question 22. The expression $\frac{2-\sqrt{k}}{1+\sqrt{k}}$ may be written in the form $a+b\sqrt{k}$. Find the values of a and b in terms of k.

Question 23.

a) Simplify $\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}}$.

b) Hence write down the solution to $\frac{\sqrt{7}-\sqrt{6}}{\sqrt{7}+\sqrt{6}}$.

Question 24. Evaluate the sum $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \cdots + \frac{1}{\sqrt{99}+\sqrt{100}}$.

Question 25. By using algebraic methods, show that $\sqrt{2}+\sqrt{3}>\sqrt{5}$.

Question 26. Show that $\frac{2}{\sqrt{5}+\sqrt{2}+1}=3+2\sqrt{2}-\sqrt{5}-\sqrt{10}$.